BCC Superalloys

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1. Introduction

While Ni-based superalloys exhibit high temperature creep resistance through coherent γ/γ' phases, superior thermal resistance materials can elevate efficiency for various combustion applications such as turbine blades or aero engines. The common problem with the strategy of γ/γ' microstructures in FCC superalloys is the solvus temperature of the γ' phase which limits operating to a max of 1100°C. Additionally, Ni melts at 1445°C, considerably lower than refractory metals (W, Mo, Nb, Ta, Re) which primarily possess in BCC structure. BCC-superalloys are nascent class of materials based on refractory BCC with ordered precipitate forming elements (Al, Ti, Zr) [1]. The objective is to create a microstructure analogous to the γ/γ' of Nickel-based superalloys, but with significant higher solvus temperature [2].

Formation of the BCC+B2 microstructure also demonstrates several challenges. First, in traditional alloys, it is challenging to synthesize thermally stable ordered precipitates [3]. Thus, utilization of complex concentrated alloys is being explored; for instance, B2-HfRu has been shown to withstand temperatures up to 1600°C. Secondly, high ductile-brittle transition temperature (DBTT) is a major concern regarding toughness of BCC-HEAs [4]. This is because superior creep resistance at high temperature translates to dislocation hindering, which inevitably limits deformability, especially at low temperature where thermal activated processes are sluggish. Finally, casting Ru containing alloys usually results in pronounced segregations while addictive manufacturing leads to cracks [5]. Simultaneously, the BCC structure has higher diffusivity due to low atomic packing factor, causing severe oxidation and weight loss during elevated temperature operation [6].

Despite these processing and ambient ductility challenges, the potential for BCC-superalloys to exceed the fundamental thermodynamic limits of Ni-based systems has driven significant research momentum. Consequently, a comprehensive understanding of recent alloy design strategies is essential for future development. This review aims to provide essential point of view about the BCC-superalloys stage of research, We start by reintroducing the concepts of superalloys and dislocation mechanisms distinguishing BCC from FCC systems. Then, have a review of Fe-based BCC superalloys along with crucial factors, specifically lattice misfit, precipitations and how they influence overall creep resistance. Next, we focus on HEA materials's microstructure control, deformation mechanisms and oxidation issues. Finally, we summarize current limitations of HEA and the manufacturing strategies to produce desired structures.

2. Fundamental Physics: Crystal Structure Dictates Behavior

Body-centered cubic (BCC) superalloys represent a promising, yet challenging, alternative to the well-established face-centered cubic (FCC) nickel-based superalloys. While FCC Ni-based superalloys have dominated high-temperature structural applications for decades, BCC systems—particularly Fe-based ferritic superalloys—offer advantages including lower density, reduced raw material costs, superior thermal conductivity, and lower thermal expansion coefficients. These properties make them become attractive candidates for the next-generation of ultra-supercritical power plants aiming to achieve operating temperatures above 700°C [3]. However, the crystal structure of BCC superalloys gives rise to fundamental challenges, from atomic-scale dislocation core structures to macroscopic mechanical behavior. Therefore, comprehending these interconnected phenomena is essential for advancing BCC superalloy development, since only with such insight the alloy design can be guided in a more effective direction [1].

2.1. Topological Differences and Intrinsic Lattice Resistance

The fundamental difference in the mechanical behavior of FCC and BCC superalloys is based on their atomistic topology of the dislocation cores. This topology dictates the magnitude of the Peierls stress (τ_p), the intrinsic lattice friction resisting dislocation glide and subsequently governs the temperature dependence of yield strength and the ductile-to-brittle transition [2].

In FCC systems, the high atomic packing factor (≈ 0.74) facilitates deformation along the close-packed $\{111\}$ planes. The energetics of the perfect lattice dislocation, $a/2\langle 110\rangle$, favor dissociation into two Shockley partials separated by an Intrinsic Stacking Fault (ISF) according to the reaction:

$$a/2\langle 110\rangle \rightarrow a/6\langle 211\rangle + \text{ISF} + a/6\langle 121\rangle$$

This dissociation creates a two-dimensional planar core that significantly lowers the Peierls stress by lattice distortion, while simultaneously pinning the dislocation to a particular glide plane. This planar restriction governs the system's dynamic recovery, as the transfer to an intersecting plane (cross-slip) requires the separated partials to energetically constrict back into a compact core. This requirement creates a fundamental difference in behavior based on dislocation character: screw dislocations, which typically possess narrower dissociation widths, can undergo constriction to facilitate cross-slip, a process enhanced in high Stacking Fault Energy (SFE) alloys, where partial separation is naturally minimized. Conversely, edge dislocations are geometrically strictly confined to their primary glide planes. Being unable to cross-slip, edge components accumulate into stable dipoles and pile up, thereby causing the metal to get stronger [7].

In contrast to FCC systems, body-centered cubic (BCC) lattices possess a more open structure (APF ≈ 0.68) lacking true close-packed planes. This structural difference results in a fundamental distinction in dislocation behavior. While edge dislocations in BCC retain planar cores and high mobility, screw dislocations with Burgers vector $b = a/2\langle 111\rangle$ exhibit a distinct non-planar core structure [2]. Due to crystal symmetry, the screw dislocation core spreads simultaneously onto three intersecting $\{110\}$ planes along the $\langle 111\rangle$ zone axis.This threefold symmetry effectively locks the dislocation at 0 K, creating a high intrinsic lattice resistance. Consequently, the motion of the non-planar core requires thermal activation, typically proceeding via the kink-pair mechanism. Thermal fluctuations provide sufficient energy for a small segment of the screw dislocation to overcome the Peierls barrier, nucleating a pair of kinks [2]. These kinks, possessing edge character and high mobility, propagate laterally along the dislocation line under applied stress. This behavior differs significantly from FCC Ni-based superalloys, which show temperature-independent yielding, with strength remaining nearly constant at low-to-moderate temperatures. Ultimately, plasticity in BCC systems is screw-dislocation controlled. Under stress, mobile edge segments expand rapidly and exit the crystal or pile up at grain boundaries, leaving the sluggish screw dislocations behind. The macroscopic yield strength is therefore controlled by the stress required to move these screw segments via kink-pair nucleation [1]. Furthermore, the ratio of screw-to-edge dislocation mobility (v_s/v_e) dictates the Ductile-to-Brittle Transition (DBT) [1]. When this ratio drops below a critical value, dislocation sources cannot maintain multiplication, leading to catastrophic brittle failure. This mechanism explains the high Brittle-to-Ductile Transition Temperatures (BDTT) observed in BCC ferritic superalloys, limiting their room-temperature applications.

2.2. Precipitation Physics in Fe-based Ferritic Superalloys

The strengthening behavior of superalloys is governed by the thermodynamics and morphology of their intermetallic precipitates. While Nickel-based FCC superalloys rely on the $L1_2$-ordered γ' phase (Ni_3Al) distributed in a disordered γ matrix, Iron-based BCC superalloys utilize $B2$ (CsCl-type) or $L2_1$ (Heusler) ordered phases within a disordered A2 (BCC) matrix. The morphological evolution of these precipitates is driven by the competition between interfacial energy ($\Delta G_{surface}$), which minimizes surface area by favoring spheres, and elastic strain energy (ΔG_{strain}), which favors cuboidal alignment along elastically soft directions [8].

BCC superalloys can operate with a wide lattice misfit, ranging from about 0.06% to 8%. The FBB8 system (Fe-10Cr-10Ni-6.5Al-3.4Mo wt. %) are designed with low lattice misfit ($\delta < 0.5\%$). In this regime, the elastic strain energy remains negligible relative to the surface energy, regardless of the particle size. Without sufficient strain energy to drive the cuboidal transition, precipitates retain their spherical morphology (as shown in Fig. 1a) throughout the heat treatment. This minimizes the thermodynamic driving force for coarsening (Ostwald ripening), granting these alloys exceptional long-term thermal stability [4, 8]. However, this low misfit leads to a rapid loss of creep resistance at temperatures exceeding 700 $°C$ because the minimal elastic interaction between the precipitates and the matrix makes them ineffective at resisting dislocation bypass and climb. [3, 4].

To resolve the paradox of the low misfit regime, where thermal stability is gained at the expense of mechanical strength, recent alloy development has focused on Titanium optimization in ferritic superalloys. While single-phase B2 precipitates in low-misfit systems (like FBB8) offer limited resistance to dislocation bypass due to reduced elastic interactions, adding Titanium to the FBB8 systems (e.g., FBB8+2Ti) significantly modifies the lattice parameters. In this alloy, the addition of 2 wt. % Ti leads to the formation of precipitates with a lamellar morphology with the alternative arrangement of B2 and $L2_1$, as shown in Fig. 1b. This phenomenon could be related to the spinodal decomposition of the original B2 phase into B2 and $L2_1$ phases caused by Ti addition. As the Ti content increases, the B2 phase is gradually ordered and finally transforms completely into the $L2_1$ phase, as shown in Fig. 1c. This addition increases the lattice misfit from the near-zero level of the base alloy 0.06% to a range of $0.67\% - 1.39\%$ [3, 4].

This thermodynamic adjustment drives the formation of a unique hierarchical microstructure. Ti enters to the B2 phase, creating an ordered arrangement between Al and Ti atoms on the Al sublattice. This ordering leads to the formation of the $L2_1- Ni_2TiAl$ structure (Fig. 2). The precipitates evolve into complex multi-scale structures consisting of nanoscale $L2_1-$ Ni_2TiAl domains (5–20 nm) embedded within larger parent $B2$-NiAl precipitates (50–200 nm). This microstructure transforms the simple Orowan barrier of single-phase alloys into a complex, dual-barrier energy landscape for dislocations [3]. The increased lattice misfit of B2 strucutre reintroduces a moderate level of coherency strain, producing stronger elastic interactions that hinder dislocation climb and bypass. When a dislocation breach the outer shell, it encounters the internal $L2_1$ domains. The high order-hardening of the Heusler phase acts as a rigid, multi-layered barrier, resisting cutting even at high stresses. This strategy successfully balances the trade-off between microstructural stability and strength [3, 4]. The misfit remains moderate enough to maintain coherency, while the hierarchical barriers significantly enhance mechanical performance. Experimentally, Ti-optimized alloys (e.g., FBB8+2Ti) achieve a $\sim 35\%$ increase in room-temperature yield strength (from 917 MPa to 1236 MPa). In terms of creep resistance, the threshold stress at 973 K (700°C) improves from 69 MPa in the base alloy to 179 MPa in the Ti-optimized variant with minimum creep rates dropping to the order of $\sim 10^{-8}$ s^{-1} under high stress conditions [4]. However, excessive Titanium addition (e.g., 4%) must be avoided, as it can lead to excessive misfit ($\delta \approx 0.5\%$) and complete transformation to $L2_1$ which may actually reduce the creep threshold stress due to the loss of coherency [3, 4].

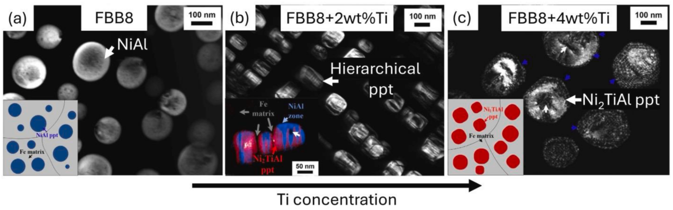

Figure 1: TEM images of FBB8 alloy with different Ti additions showing the evolution of microstructure. Samples were reated at 1200 °C for 0.5 h and aged at 700 °C for 100 h, followed by air cooling after each treatment. Crystal structures of disordered BCC, NiAl-B2 and Ni_2TiAl-$L2_1$. a) At Ti = 0, the alloy exhibits a two-phase structure of BCC-Fe and B2-NiAl, b) and c) With increasing Ti content, the $L2_1$-Ni_2TiAl is gradually formed, resulting in a three-phase (BCC-Fe + $L2_1$-Ni_2TiAl) at Ti = 2-4 wt.% [3]

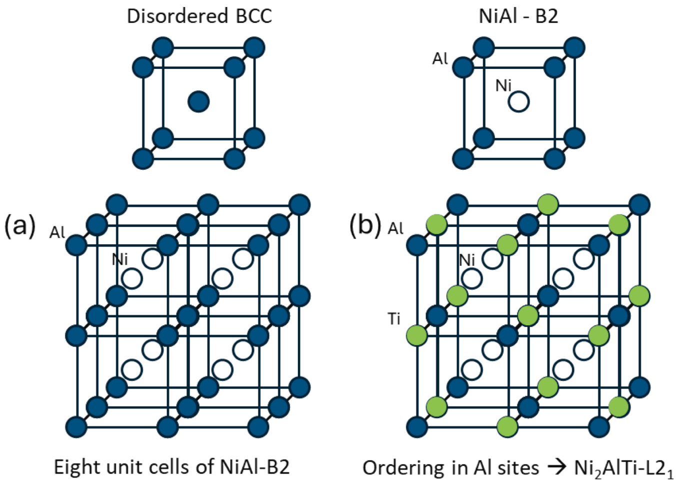

Figure 2: Crystal structures of disordered BCC, NiAl-B2 and Ni_2TiAl-$L2_1$. a) Eight unit cells of B2-NiAl structure, b) ordering on Al sites of B2 to result in $L2_1$-Ni_2TiAl. [3]

2.3. Strengthening Mechanisms and Dislocation-Precipitate Interactions

The yield strength of precipitation-hardened superalloys is fundamentally governed by the interaction between moving dislocations and the precipitates. When a dislocation encounters a precipitate, the mechanism is determined by a competition between shearing (cutting through the particle) and Orowan looping (bypassing the particle). The transition depends on the particle's size relative to a critical radius (r_c), defined as the point where the stress required for shearing equals that for looping ($\Delta\sigma_{shear} = \Delta\sigma_{Orowan}$). Below this critical size, shearing dominates; above it, Orowan looping prevails [4].

In conventional FCC superalloys reinforced by $L1_2$ precipitates, such as Ni_3Al, a dislocation attempting to cut through the ordered secondary phase and creates a region of local disorder known as an Anti-Phase Boundary (APB). The energy required to create this boundary acts as a significant barrier to dislocation motion, resulting in what is known as order strengthening. To minimize the energy associated with forming these boundaries, dislocations move through the ordered precipitates in pairs of super-dislocations. Within these pairs, the trailing dislocation eliminates the APB produced by the leading dislocation, allowing the pair to glide more efficiently than a single dislocation [2]. Critically, at elevated temperatures, the screw superpartials of these dislocations can thermally cross-slip from their close-packed {111} glide planes onto {010} planes. This forms immobile non-planar configurations known as Kear-Wilsdorf locks, which act as barriers to further slip. This mechanism is responsible for the Yield Strength Anomaly (YSA), where the strength of $L1_2$-strengthened alloys paradoxically increases with temperature up to approximately 800°C [9].

In B2-reinforced BCC superalloys, the deformation mechanism is fundamentally dictated by the ordering energy of the intermetallic phase. For ferritic superalloys utilizing stoichiometric B2 phases, the high Anti-Phase Boundary (APB) energy creates a significant barrier to particle shearing [2]. This high APB energy restricts shearing to at very small particle sizes, such as the 1.5–3.1 nm range observed in underaged Fe-Ni-Al-Cr alloys [4]. Consequently, for microstructures with typical precipitate sizes (>30 nm), dislocation motion is governed by the Orowan bypassing mechanism, where the Orowan equation for the increase in critical resolved shear stress $\Delta\tau$ is defined as:

$$\Delta\tau = \frac{2T}{b(L - 2r)}$$

where $\Delta\tau$ is representing the strengthening stress (specifically the increase in shear stress required to move a dislocation), T is thhe line tension of the dislocation (the energy required to bow the dislocation line), b is the Burgers vector, and $(L - 2r)$ represents the free distance or effective inter-particle spacing λ between hard precipitates of radius r [8]. Because the stress $\Delta\tau$ is inversely proportional to the inter-particle spacing $(L - 2r)$, the mechanical behavior of BCC superalloys is highly sensitive to microstructural coarsening. In these systems, the strengthening mechanism transitions from particle shearing to Orowan bypassing as precipitates grow past a critical radius (r_0). In ferritic alloys (e.g., Fe-Ni-Al-Cr-Mo), particles as small as 1.5–3.1 nm are sheared, while larger precipitates (>54 nm) trigger the Orowan looping mechanism [4]. As precipitates coarsen and the spacing (λ) increases, strength decreases. For instance, in an Fe-10Cr-10Ni-6.5Al-3.4Mo , the yield strength is high in the peak-aged condition ($\sim$ 917 MPa at room temperature) but drops significantly at elevated temperatures (e.g., $\sim$ 120 MPa at 973 K) as thermal activation and coarsening facilitate dislocation climb and bypass [3]. Unlike FCC superalloys where Kear-Wilsdorf locking creates a Yield Strength Anomaly (YSA) up to intermediate temperatures, the strength of BCC superalloys generally decreases monotonically with temperature, as deformation is controlled by the high lattice friction (Peierls stress) of screw dislocations in the BCC matrix. These fundamental differences dictate specific strategies for alloy design. To maximize strength via the Orowan mechanism, the microstructure must be optimized to minimize inter-particle spacing. While increasing the volume fraction of precipitates theoretically enhances strength, it presents a critical trade-off in ferritic systems: exceeding a volume fraction of $\sim 10\%$ can cause room-temperature ductility to drop dramatically to $< 1\%$, resulting in brittle intergranular fracture. In contrast, Refractory High-Entropy Alloys (RHEAs) can sustain much higher precipitate fractions (e.g., basket-weave microstructures) to achieve ultra-high strength (up to $\sim$ 2000 MPa), but often at the cost of ductility [3]. Therefore, the successful design of BCC superalloys necessitates a careful balance between achieving peak strength through refinement (r≈ 40-60 nm and maintaining that strength through kinetic stabilization (e.g., using slow-diffusing elements or hierarchical structures) to prevent rapid coarsening at service temperatures [4].

2.4. Kinetic Stability: The Diffusion Challenge

The high-temperature performance of superalloys is fundamentally limited by atomic diffusivity, a property dictated by the crystal lattice structure. The Body-Centered Cubic lattice, with an Atomic Packing Factor (APF) of 0.68, possesses a significantly more open structure than the close-packed Face-Centered Cubic lattice (APF = 0.74). This structural openness lowers the activation energy for vacancy migration and increases the number of available interstitial sites. Consequently, at equivalent homologous temperatures, self-diffusion coefficients in BCC metals are typically higher than in their FCC counterparts. This rapid diffusion acts as a primary driver for degradation at elevated temperatures, contributing to increased creep rates and precipitate coarsening. This difference in diffusivity leads to distinct deformation challenges. While FCC superalloys rely on the stability of the γ' phase, BCC superalloys (such as ferritic alloys strengthened by B2-NiAl) suffer from a rapid loss of creep resistance at temperatures above $700°C$ [3]. Contrary to the viscous glide mechanism observed in some solid solution alloys, the degradation in these precipitation-hardened BCC systems is primarily driven by dislocation climb. The fast lattice diffusion facilitates the climb of dislocations over precipitates, while the relatively low lattice misfit in systems like FBB8 offers limited elastic resistance to this bypassing mechanism [4]. Additionally, deformation in the BCC matrix is strongly influenced by high lattice friction (Peierls stress) and the motion of screw dislocations with non-planar cores, distinguishing it from the planar slip often seen in FCC systems. The most critical implication of the high diffusivity in BCC systems is the accelerated coarsening and microstructural instability of strengthening precipitates. In ferritic superalloys, thermal aging can lead to rapid coarsening or phase transformations (e.g., B2 to $L2_1$) that result in a loss of coherency. For instance, in the FBB8 system, while precipitates may be spherical and coherent initially (r ≈ 65 nm) [3], high-temperature exposure facilitates diffusion-assisted processes that degrade the threshold stress necessary to prevent creep.

To mitigate the kinetic disadvantages of the open BCC structure, alloy design strategies must focus on slowing diffusion or stabilizing the coherent microstructure. RHEAs leverage elements with high melting points (e.g., W, Mo, Nb) to extend the absolute service temperature range to $1300-1800°C$ [3]. However, rather than simply bypassing diffusion issues, these alloys must combat the inherently high atomic diffusivity of the BCC lattice, which promotes dislocation climb and creep at these elevated temperatures. In ferritic superalloys, kinetic stabilization is achieved not just by refractory alloying, but by engineering hierarchical microstructures. For example, adding Ti to Fe-Ni-Al systems promotes the formation of $L2_1$ - Ni_2TiAl domains within the B2 precipitates [3]. This hierarchical architecture increases the lattice misfit and introduces stronger elastic interactions and higher Peierls stresses, which effectively hinder dislocation climb and bypass mechanisms. Furthermore, the complex atomic environment inherent to multi-principal element systems contributes to microstructural stability. The presence of chemical short-range ordering (SRO) creates a "sluggish diffusion" effect, which effectively retards elemental diffusion and slows down the coarsening of coherent precipitates [4]. Consequently, the combination of hierarchical barriers, such as $L2_1$ domains integrated with B2 phases, and the sluggish kinetics of high-entropy matrices acts to delay the onset of coarsening and preserve mechanical integrity at service temperatures.

3. High Entropy Alloys: From Microstructures Mechanics to Engineering

The development of RHEAs represents a major paradigm change in the science of high-temperature materials science. Unlike traditional superalloys, RHEAs utilize a BCC matrix. This fundamental difference in crystal structure strongly influences the intrinsic deformation mechanisms of the material, requiring new approaches to alloy design. The mechanical performance of these alloys is defined by a competition between the intrinsic lattice friction of the disordered complex matrix and the ordered constraints created by B2 precipitates [1].

In BCC refractory alloys, plastic deformation is mainly governed by the movement of screw dislocations. Due to the non-planar core structure spreading onto multiple intersecting slip planes, screw dislocations in BCC metals naturally experience Peierls stress differently from edge dislocations. This turns kink-pairs into the main displacement mechanism of screw dislocations. However, the physics of RHEAs extends beyond simple lattice distortion. The assumption of a random solid solution is increasingly challenged by the presence of Local Chemical Order (LCO) or Short-Range Order. These non-random atomic clusters create spatially varying chemical friction that acts as extrinsic pinning points for dislocations, effectively roughening the Pierls potential. Therefore, screw dislocations tend to remain straight and confined to the core during deformation. It becomes energetically favorable for dislocations to shear the precipitates than to bow into the high friction matrix. Consequently, the critical radius r_c in RHEAs is shifted to larger values compared to conventional alloys [2].

3.1. Deformation Mechanisms

Due to the rugged matrix potential, r_c is shifted to larger values (experimentally observed > 50nm in RHEAs vs < 20nm in Ni-superalloys). Maintaining precipitates within an optimized scale (tyopically 50-100 nm) is critical to exploit the extended shearing regime [2]. When a matrix dislocation enters an ordered B2 precipitate, it disrupts the ordered B2/$L2_1$ superlattice, creating a high energy APB defect. In RHEAs, the high configurational entropy of the precipitate phase lowers the APB energy (γ_{APB}). This reduction in APB energy allows super-dislocations to dissociate into widely separated partials, enabling the activation of ductile <111> slip even within the ordered lattice. Experimental observations confirm this in BCC alloys like $AlMo_{0.5}NbTa_{0.5}TiZr$, where paired dislocations are seen cutting through coherent precipitates via the Entropy-Stabilized Ductility mechanism [2].

The diffusion dynamics is controversially separated into distinct temperature regimes. At intermediate and high temperatures ($T > 0.6T_m$), thermal energy allows dislocations to escape their slip planes through climbing mechanism, a process driven by vacancy diffusion. While the BCC matrix is often assumed to have sluggish diffusion, recent studies indicate that vacancy immigration in the disordered matrix can be surprisingly rapid, compared to conventional alloys in relative scale [10]. While the matrix behave as high-mobility conduit for dislocations, the creep rate remains low because the B2 precipitate effectively block the movement of defects. Diffusion within the ordered lattice requires complex, cooperative sublattice jumps to disrupt the energy bonding. This process is significantly slower than diffusion in the matrix. The overall creep rate is thus not determined by the average diffusivity of the alloy, but is limited by the slower diffusion required for dislocations to climb over the ordered phase. An example for B2 phase diffusion hindering is $Ta_{27.3}Mo_{27.3}Ti_{27.3}Cr_8Al_{10}$ which exhibits minimum creep rates ($2.5 * 10^{-9} s^{-1}$) comparable to CMSX-4-single-crystal superalloys at 1273 K, despite the inherent faster diffusion rate within the BCC lattice [4].

3.2. Microstructural Architecture: Morphology, Size, and Volume

While the deformation mechanisms dictate single dislocation behavior, bulk mechanical performance requires the microstructural architecture to enforce dislocation and precipitate interaction. To fully grasp the precipitate design, three major critical factors are considered: morphology, lattice misfit, and volume fraction.

For high-temperature creep resistance, a coherent cuboidal morphology is strictly superior to various shapes. A cuboidal precipitate presents flat 100 facets to incoming dislocations, maximizing the path length a dislocation must shear and efficiently form narrow, uniform matrix channels, which constraint dislocation segments. The morphology is controlled by lattice misfit δ between the precipitate and the matrix [8]. A moderate misfit (typically between 0.5% and 1.0%) creates the competition between strain and interfacial energy which results in cuboidal shape. A low misfit (near 0%) translates into spherical precipitates providing thermodynamic stability whereas a δ above 2% causes the lattice to be incoherent, leading to the formation of semi-coherent plates or basket-weave structures that worsen mechanical properties at elevated temperature. Directional stability is governed by rafting, which depends on the sign of the misfit [8]. In P-Type (positive lattice misfit $\delta > 0$) cuboidal precipitates form connection parallel to the loading direction, creates open channels for dislocation gliding and accelerating creep failure. In contrast, N-Type (negative misfit, $\delta < 0$) promotes rafting perpendicular to the stress axis, effectively halting dislocation movement by forming continuous barriers. Furthermore, the alloy's strength and ductility are defined by the interplay between precipitate volume and topology. While increasing the volume fraction of precipitates theoretically enhances strength—allowing RHEAs to sustain higher fractions to achieve ultra-high strength—it presents a critical topological trade-off. Increasing the volume fraction in some compositions results in the inversion of the matrix and precipitates phase. A clear example is the $AlMo_{0.5}NbTa_{0.5}TiZr$ alloy Unlike Ni-based alloys where the matrix remains continuous even at high precipitate fractions, BCC-HEAs often undergo spinodal decomposition where the phase with the larger volume fraction spontaneously becomes the continuous matrix [3]. The inverted matrix forms a continuous rigid B2 network with isolated ductile BCC islands. The phenomena is catastrophic for ductility, as cracks propagate rapidly through the brittle continuous phase.

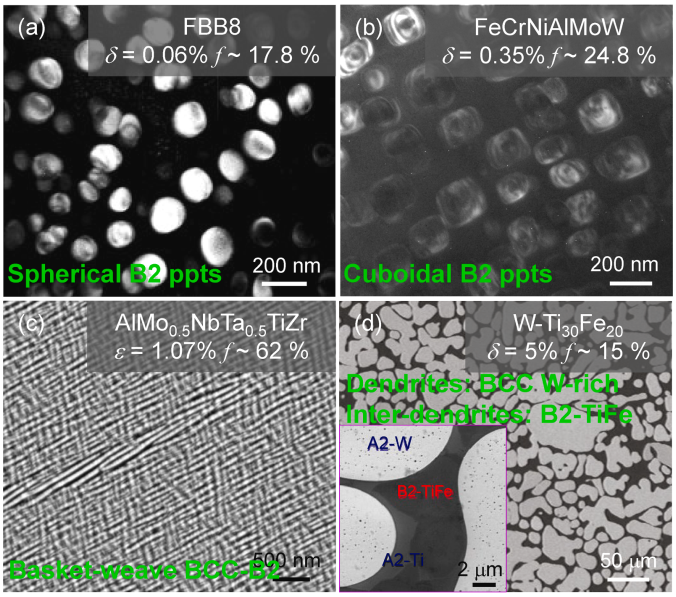

Figure 3: Typical coherent between BCC matrix and B2 precipitates a) small misfit lead to spherical morphology b) moderate misfit with cuboidal representation c) and d) large misfit, adapted from [4]

3.3. Strategies for Degisn

The successful deployment of BCC-superalloys requires a holistic design strategy that integrates thermodynamic stability, microstructure control, and environmental resistance. While the theoretical potential of these alloys is vast, navigating through complex manifold of transformation pathways and realizing the trade-off between mechanical properties are the real challenges.

The natural idea is to modify compositions and control their evolution. For instance, Al is a crucial component in RHEAs, yet it is directly involved in a major engineering issue known as microstructural inversion (or topological inversion). To resolve this, researchers have developed strategies to manipulate the phase transformation pathway from spinodal decomposition to nucleation and growth [3]. A success attempt is $Al_{10}Nb_{15}Ta_5Ti_{30}Zr_{40}$ where lowering the ordering temperature (T_c) ensures the ductile BCC phase solidifies first, forming a continuous matrix before the B2 phase precipitates, thus restoring ductility. Another example is the use of addictive manufacturing to control solidification. Deleterious phases such as Laves (C14, C15) and Omega (ω) are undesired but exist in traditional casting due to slow cooling rates. Additive manufacturing has the advantage of rapid cooling ($10^3 - 10^5 K/s$) which can kinetically bypass these brittle intermetallics. Furthermore, the addition of Ru significantly improve stability of B2 phase. However, Ru-containing alloys exhibit a widened solidification range, making them susceptible to solidification cracking [5]. Design parameters must therefore balance refractory content with melt pool fluidity to prevent hot tearing.

Alloying for oxidation resistance presents a critical trade-off in RHEA design. RHEAs are opened to rapid oxidation and mass loss. While high Al content is necessary to form protective alumina (Al_2O_3) interface, it often increase lattice misfit and the material exhibit lower ductility. A critical design strategy involves mitigating pest oxidation in the intermediate temperature range ($600 - 900°C$) to suppress the formation of deleterious oxides such as volatile MoO_3 or liquid-forming V_2O_5. [6].

Finally, low ductility and crack mitigation remain challenges. grain boundary engineering is essential. The segregation of interstitial elements like B or C can enhance boundary cohesion by changing the fracture mode from intergranular to transgranular. Furthermore, promoting a dual-phase microstructure where a softer BCC phase acts as a crack blunter can provide extrinsic toughening, arresting cracks initiated in the harder B2 phase [4].

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Appendix A. Summarize of Chemical elements effects & Example BCC-RHEAs

Element	Detailed Effects on Microstructure & Mechanics
	Precipitate Stabilizers & Density Modifiers
Aluminum (Al)	Primary B2-former essential for ordered precipitation hardening[1]. Reduces alloy density. Critical for high-temperature oxidation resistance via formation of protective Al_2O_3 scales[6]. **Risk:** High concentrations drive topological phase inversion, creating a brittle continuous B2 matrix[4].
Titanium (Ti)	Enhances room-temperature ductility and lowers density[1]. Modifies the ordering temperature (T_c) of the B2 phase. Acts as a stabilizer for the BCC solid solution in certain compositional ranges to prevent premature intermetallic formation[3].
Ruthenium (Ru)	Extends the stability window of ordered B2 precipitates to ultra-high temperatures ($> 1200°C$)[5]. **Risk:** Significantly widens the solidification freezing range, increasing susceptibility to hot tearing (solidification cracking), particularly in Additive Manufacturing[5].
	Refractory Matrix Base
Niobium (Nb)	Fundamental matrix builder providing high melting point (T_m) and solid solution capacity. Forms the continuous ductile channels necessary for dislocation gliding in the matrix[1].
Tantalum (Ta)	Significantly increases creep resistance due to large atomic mass and slow diffusion kinetics[4]. Increases T_m and elastic modulus but raises density.
Molybdenum (Mo)	Potent lattice misfit (δ) tuner; used to adjust the lattice parameter to achieve optimal cuboidal precipitate morphology (negative or positive misfit)[8]. **Risk:** Susceptible to catastrophic "pesting" oxidation due to volatile MoO_3 formation[6].
	Process & Environmental Control
Zirconium (Zr)	Strong B2 stabilizer. **Risk:** Partitions heavily to the liquid phase during solidification, creating wide freezing ranges ($> 450°C$) that lead to severe solidification cracking[5]. Promotes deleterious Laves phase formation if not controlled [3].
Hafnium (Hf)	Used as a substitute for Zr to mitigate solidification cracking[5]. Narrows the freezing range and reduces segregation-induced hot tearing in Additive Manufacturing processes.
Chromium (Cr)	Critical for environmental defense. Mitigates "pesting" oxidation in the intermediate temperature range ($600 - 900°C$)[6]. Reduces the lattice parameter of the BCC matrix, influencing the misfit sign[8].
Silicon (Si)	Synergizes with Cr and Al to promote stable scale formation and suppress volatile oxide evaporation[6].
B / C (Interstitials)	Grain boundary engineers. Segregate to grain boundaries to enhance cohesion and suppress intergranular fracture modes, improving overall ductility[4].

Table A.1: Summary of Chemical Elements and their effects to BCC-RHEAs

Alloy System	Key Properties & Microstructural Features
	The Ductility-Inversion Class
AlMo0.5NbTa0.5TiZr	**Entropy-Stabilized Ductility:** Exhibits ductile behavior via partial dislocation dissociation and planar slip, despite ordered precipitates[2]. **Topology Trap:** Highly susceptible to phase inversion, forming a brittle continuous B2 matrix with isolated ductile BCC islands, leading to premature fracture[4].
Al10Nb15Ta5Ti30Zr40	**Pathway Engineered:** Designed to solidify via Nucleation and Growth rather than Spinodal Decomposition. Lowering the ordering temperature (T_c) forces the ductile BCC phase to solidify first, creating a continuous ductile matrix and restoring mechanical stability[3].
	The Creep-Resistant Class
Ta27.3Mo27.3Ti27.3Cr8Al10	**The Diffusion Barrier:** Demonstrates minimum creep rates ($2.5 \times 10^{-9} s^{-1}$) at 1273 K, comparable to single-crystal CMSX-4 Ni-superalloys[4]. The B2 precipitates act as effective diffusion barriers despite the faster intrinsic diffusion of the BCC matrix.
	The High-Temperature & AM Class
Ru-containing Refractories	**Ultra-High T Stability:** Maintains stable, ordered B2 precipitates at temperatures exceeding 1200°C, significantly higher than traditional Ni-based superalloys[5].
Zr-containing Alloys	**Solidification Cracking:** Exhibits severe "hot tearing" during Additive Manufacturing due to Zr partitioning to the liquid, resulting in wide freezing ranges (> 450°C)[5].
Hf-substituted (e.g., Hf_5Ru_4)	**AM Compatible:** Substitution of Zr with Hf narrows the freezing range, successfully mitigating solidification cracking and enabling defect-free 3D printed components[5].
	The Environmental Defense Class
Al-Cr-Si Doped Systems	**Pesting Suppression:** Specifically formulated to resist catastrophic oxidation in the $600 - 900$°C "pesting" regime by suppressing the formation and evaporation of volatile oxides like MoO_3 and V_2O_5[6].

Table A.2: Key BCC-HEAs from the References